

DevAlpha Technologies
Civil Engineering & Structural Design Internship

Task 2: Building Design Planning

Residential Building - Double Storey Floor Plan Documentation

1. Project Title & Objective

Project Title: Residential Building Design Plan - Double Storey Layout

Objective: To prepare a complete architectural and structural design plan for a double-storey residential building, including a column-beam structural grid, room layout, doors and windows, and supporting design documentation in line with standard construction design planning practice.

2. Building Parameters

The table below summarizes the key design parameters and assumptions adopted for this project.

Project Title	Residential Building Design Plan
Building Type	Double-storey residential building (G+1)
Number of Floors	2 (Ground Floor + First Floor)
Structural System	RCC framed structure (column-beam grid)
Column Grid Spacing	12'-6" horizontal x 13'-6" vertical (same grid for both floors)
Column Size (assumed)	230 mm x 305 mm (9" x 12") - uniform for both floors
Beam Size (assumed)	230 mm x 305 mm (9" x 12"), connecting all columns
Wall Thickness (external)	230 mm (9")
Wall Thickness (internal)	115 mm (4.5")
Slab Thickness (assumed)	125 mm
Total Carpet Area	~1,070 sq-ft per floor (~2,140 sq-ft for both floors)
Approx. Built-up Area	~1,250-1,300 sq-ft per floor (~2,500-2,600 sq-ft for both floors)
Material Grade (assumed)	Concrete M25, Steel Fe415

3. Design Approach & Methodology

The design process followed for this project is summarized below:

- Established a structural column grid with a bay spacing of 12'-6" (horizontal) x 13'-6" (vertical), forming a 3 x 2 bay layout, identical for the Ground Floor and First Floor.
- Connected columns with 9" x 12" beams to define the structural floor framing, repeated identically at both floor levels.
- Overlaid the architectural floor plan on the structural grid, defining rooms, walls, and circulation spaces.
- Assigned wall thicknesses for external (load-bearing) and internal (partition) walls.
- Added doors and windows to all habitable rooms for access, ventilation, and natural lighting.
- Computed room-wise carpet area and total built-up area for the plan.

4. Room Layout & Zoning

The building is organized into three functional zones to separate private, service, and common living spaces. The first-floor layout generally follows the same structural grid as the ground floor. However, the Drawing Room area has been replaced with an open terrace. Therefore, the enclosed carpet area of the first floor is reduced compared to the ground floor while maintaining the same structural framing system.

Zone	Rooms Included
Private Zone	Bed Room (1), Bed Room (2), Bath Room (1), Bath Room (2), Store Room
Service Zone	Kitchen, Staircase
Common / Living Zone	Living Room, Drawing Room

5. Room-wise Area Calculation

Room dimensions were taken from the finalized floor plan. Areas below are approximate, calculated as length x width per room (carpet area, excluding wall thickness), and represent the typical floor layout repeated at both Ground Floor and First Floor levels.

Room Name	Dimensions	Area (sq-ft)
Bath Room (1)	8'-0" x 6'-0"	48
Bed Room (1)	15'-1.5" x 13'-10.5"	209
Store Room	5'-6" x 7'-0"	38.5

Room Name	Dimensions	Area (sq-ft)
Bed Room (2)	17'-4.7" x 13'-10.5"	240
Bath Room (2)	8'-0" x 6'-0"	48
Kitchen	12'-6" x 7'-1"	88.5
Living Room	18'-1.5" x 12'-6"	226.5
Drawing Room	12'-6" x 13'-9"	172
Total Carpet Area		1070

Note: The total built-up area per floor (including wall thickness and circulation space) is estimated at approximately 1,250–1,300 sq-ft, which is typically 17–20% greater than the carpet area for residential buildings. The Ground Floor has a carpet area of approximately 1,070 sq-ft. On the First Floor, the Drawing Room has been replaced with an open terrace, reducing the enclosed carpet area by approximately 172 sq-ft and resulting in an enclosed carpet area of approximately 898 sq-ft. The structural column-beam grid remains unchanged on both floors. Room dimensions are based on the finalized floor plan and are subject to confirmation during detailed design and construction documentation.

6. Floor Plan Drawing

The finalized floor plan, including the structural grid, walls, doors, and windows, is shown below.



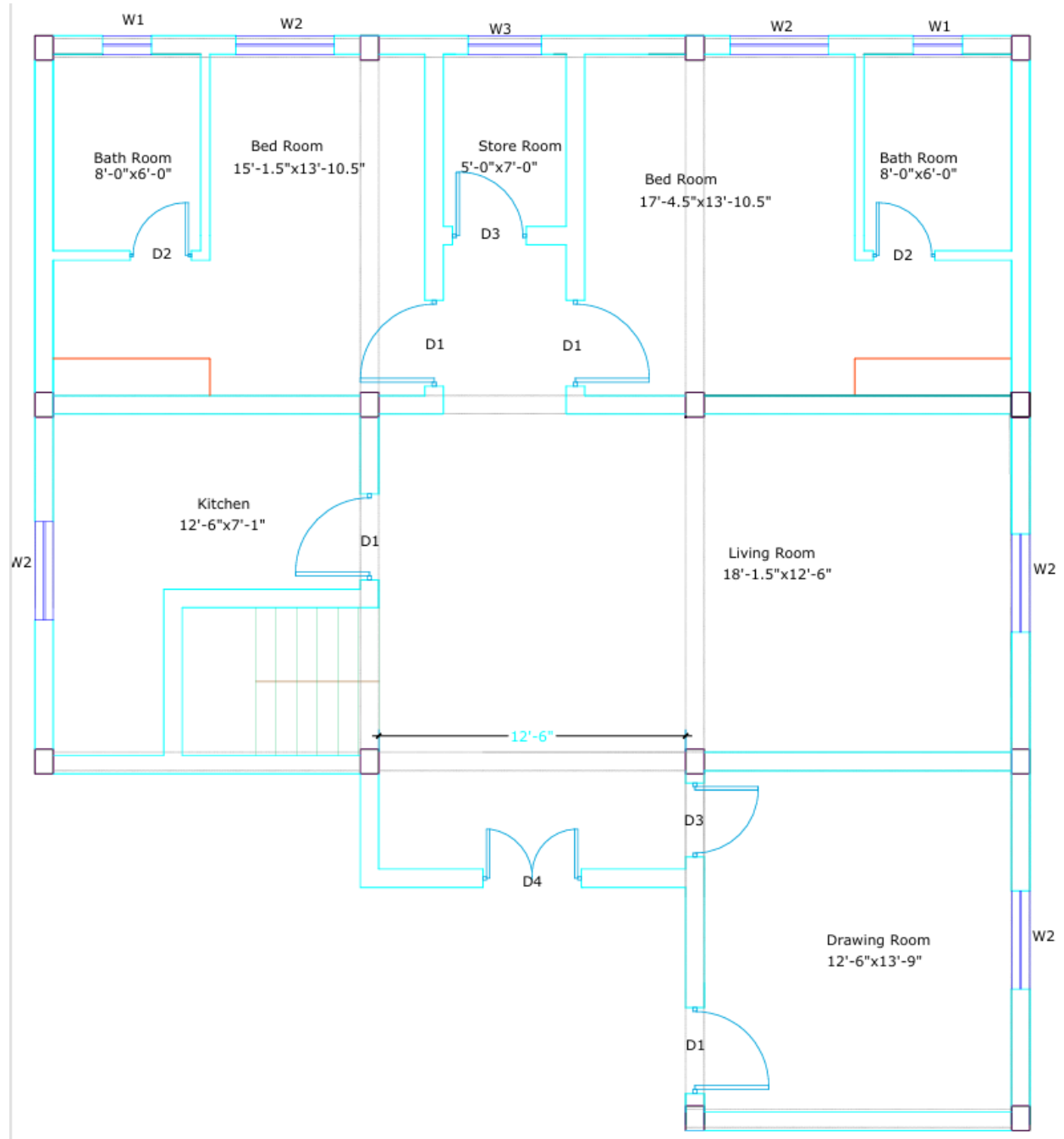


Figure 1: Typical Floor Plan (Ground Floor) - Residential Building (drawn in AutoCAD)

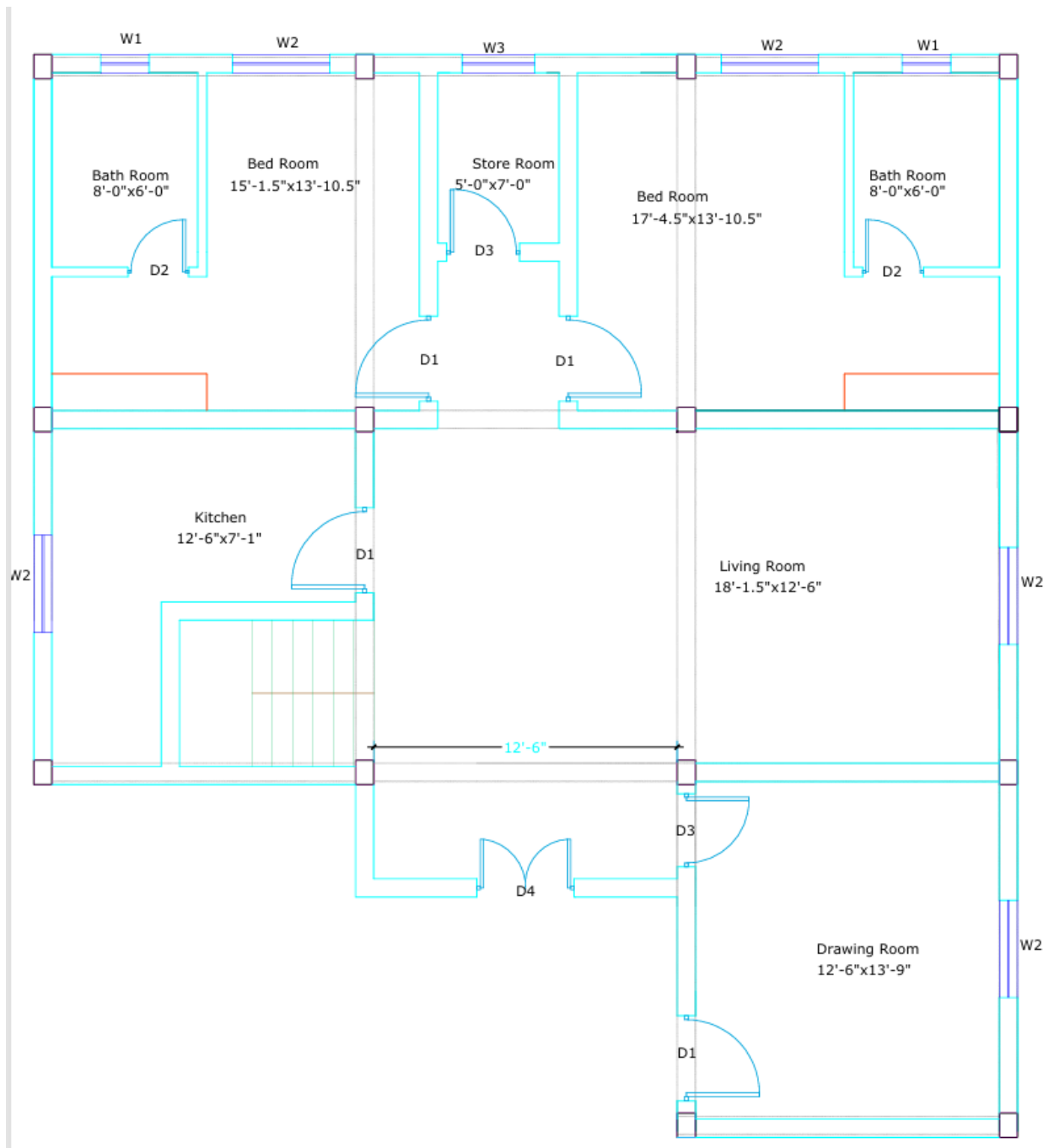


Figure 2: Typical Floor Plan (First Floor) - Residential Building (drawn in AutoCAD)

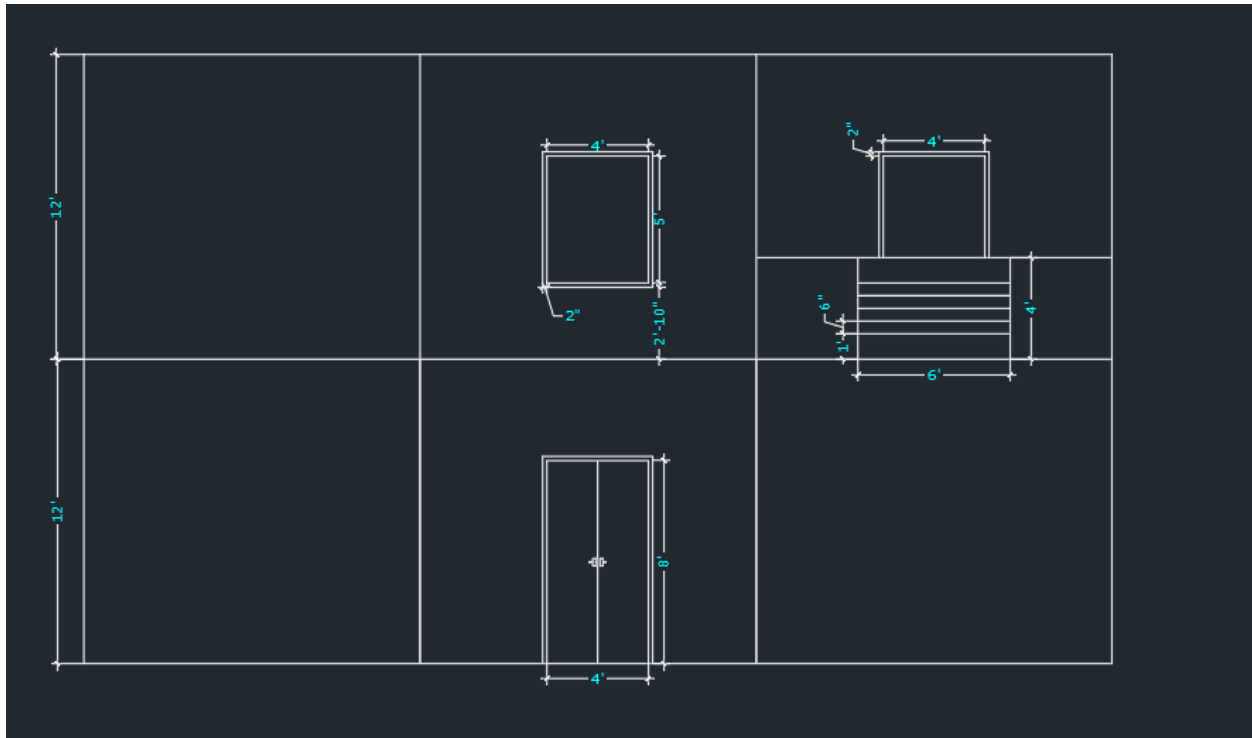


Figure 3: Typical Floor Plan Elevation - Residential Building (drawn in AutoCAD)

7. Door and Window Schedule

Finalized door sizes for the residential layout are listed below. The same door schedule applies to both the Ground Floor and First Floor, as the room layout is repeated on each level.

Door	Room / Location	Door Width
D4	Main Entrance	4'-0"
D1	Drawing Room, Kitchen, Bed Room	3'-6"
D2	Bath Room	2'-6"
D3	Store Room, Drawing Room	3'-0"

Note: The same door schedule applies to both the Ground Floor and First Floor.

Windows	Sill Level	Window Width
W1	6 ft	2'-0"
W2	3 ft	4'-0"
W3	3ft	3'-0"

Note: The same window schedule applies to both the Ground Floor and First Floor.

8. Design Assumptions

- Material grades assumed: Concrete M25, reinforcement steel Fe415 (for future structural design reference).
- The building has been developed as a double-storey (Ground Floor + First Floor) structure; the same column grid, column size, and beam size are adopted for both floors.
- Column and beam size assumed at 230 mm x 305 mm (9" x 12"), based on a bay spacing of 12'-6" (horizontal) x 13'-6" (vertical), uniform for the whole floor and repeated on both storeys.
- External wall thickness of 230 mm (9") for load-bearing walls; internal partitions at 115 mm (4.5").
- Slab thickness assumed at 125 mm, typical for residential spans of this size.
- Door sizes have been finalized as per the Door Schedule in Section 7; window sizes follow standard residential proportions and are to be finalized in the detailed design stage.
- No detailed load or safety factor calculations are included in this task; these are addressed separately under structural analysis.

9. Conclusion

This task involved preparing a complete construction design plan for a double-storey (Ground Floor + First Floor) residential building, starting from a structural column-beam grid and developing it into a fully detailed architectural floor plan with defined rooms, walls, terrace, doors, and windows. The plan achieves a functional zoning layout separating private, service, and common areas, with a total carpet area of approximately 1,070 sq-ft per floor (~2,140 sq-ft across both floors). This documentation demonstrates the design planning workflow expected at this stage of the internship and forms a suitable base for further structural analysis in subsequent project tasks.